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# Perfect Storm Labs
## Capabilities Statement

**Company:** Paul Swisher DBA FrankenBits
**Division:** Perfect Storm Labs
**Owner/Operator:** Paul Swisher
**Location:** Washington State

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## Company Overview

FrankenBits' Perfect Storm Labs specializes in advanced acoustic detection systems and AI-powered monitoring platforms for defense, research, and environmental applications. Our patent-pending technologies provide real-time detection, classification, and correlation capabilities across multiple domains.

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## Core Competencies

### Acoustic Detection & Classification
- Real-time acoustic signature detection and identification
- Multi-domain acoustic analysis (aircraft, atmospheric, seismic)
- Patent-pending acoustic detection methodologies
- Adaptive threshold triggering and anomaly detection
- Frequency spectrum analysis and pattern recognition

### Artificial Intelligence & Machine Learning
- AeroMind AI - Aircraft classification and identification system
- Neural network-based pattern recognition
- Model deployment, monitoring, and drift detection
- Training pipeline automation
- Predictive analytics and correlation engines

### Data Correlation & Validation
- Multi-source data fusion and consensus validation
- Real-time correlation of acoustic events with external data sources
- Cross-reference validation systems
- Weather and atmospheric propagation modeling
- Time-series data processing and analysis

### Enterprise Software Architecture
- Microservices-based system design
- Service-oriented architecture (SOA)
- Triumvirate control plane for distributed systems
- PostgreSQL/TimescaleDB data architecture (16 databases, 122 tables)
- Real-time data pipeline engineering
- High-availability monitoring and alerting systems

### Research & Development
- Acoustic meteorology and atmospheric monitoring
- Seismic event detection and precursor analysis
- Gravitational-acoustic coupling research
- Multi-sensor fusion and triangulation
- Advanced signal processing techniques

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## Patent Portfolio

### Issued Patent Applications

**63/928,429** - Acoustic Aircraft Detection (Filed: December 1, 2025)
Real-time acoustic signature identification and aircraft classification methodologies.

**63/949,123** - Universal Triumvirate Control Plane (Filed: December 27, 2025)
Microservices orchestration and distributed system management architecture.

**63/957,752** - Cosmic Gravitational Acoustic Resonance (Filed: January 2026)
Gravitational-acoustic detection methods with applications in seismic monitoring and cosmic event detection.

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## Technology Stack

**Languages & Frameworks:** Python, SQL, REST APIs
**Databases:** PostgreSQL 14+, TimescaleDB (time-series optimization)

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****Architecture:**** Microservices, Domain-Driven Design (DDD)
****AI/ML:**** Neural networks, supervised learning, feature engineering
****Data Processing:**** Real-time audio processing, signal filtering, FFT analysis
****Integration:**** OpenSky Network, weather APIs, astronomical data sources

Key Differentiators

â€” ****Patent-protected innovations**** - Three filed patents covering core detection methodologies
â€” ****Multi-domain expertise**** - Aircraft, atmospheric, seismic, and gravitational acoustic detection
â€” ****Validated systems**** - Real-world deployments with proven detection capabilities
â€” ****Research-grade architecture**** - Enterprise-scale data systems (16 databases, 122 tables)
â€” ****AI-enhanced detection**** - Machine learning classification with continuous improvement
â€” ****Multi-source validation**** - Consensus-based correlation across independent data sources

Applications

Defense & Security

- Aircraft detection and identification
- Airspace monitoring and violation detection
- Multi-sensor surveillance systems
- Real-time threat assessment

Environmental Monitoring

- Atmospheric acoustic analysis
- Weather system characterization
- Seismic event detection
- Environmental impact assessment

Research & Scientific

- Acoustic propagation modeling
- Gravitational-acoustic coupling studies
- Seismic precursor analysis
- Multi-domain correlation research

Past Performance

****SeismoSentinel Deployment (2026)****

- Multi-station seismic detection system
- Validated 1,800-mile detection range
- Real-time atmospheric acoustic-gravity wave monitoring

****SentinelCore Platform (2025-2026)****

- Real-time aircraft detection and correlation system
- Multi-source data validation architecture
- AeroMind AI classification engine deployment

****Acoustic Meteorology Research (2025)****

- 10+ hour weather prediction lead time validation
- Atmospheric river detection and characterization
- Novel acoustic weather analysis methodologies

Contact Information

****Perfect Storm Labs****

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****SAM.gov Registration:**** Active

****CAGE Code:**** 17KK4

****Primary NAICS Code:**** 541715 - Research and Development in the Physical, Engineering, and Life Sciences

This capabilities statement contains high-level descriptions of patent-pending technologies. Detailed technical specifications and methodologies are available under NDA during proposal evaluation phases.

****Last Updated:**** January 28, 2026